MILITARY SPECIFICATION

WATTMETERS, SWITCHBOARD

TYPE, 4-1/2 INCH

This amendment forms a part of Military Specification MIL-W-19088(SHIPS), dated 24 October 1955.

PAGE 1

1.2.1.2.2, line 3: Delete "260" and substitute "250".

PAGE 2

1.2.1.9.1, line 1: Delete "MR491W300AC, KWH" and substitute "MR491W300-ACKWH".

PAGES 2 AND 3

2.1: Delete reference to Military Specification "MIL-I-15024", Military Standard "MIL-STD-105", and Navy Department Specification "General Specifications for Inspection of Material", and add the following Military Specification and Standard:

SPECIFICATION

MILITARY
"MIL-E-15090 - Enamel Equipment, Light-Gray (Formula No. 111)."

STANDARDS

MILITARY
"MIL-STD-130 - Identification Marking of U. S. Military Property."

PAGE 4

3.1: Delete and substitute:

"3.1 Qualification.- Wattmeters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3)."

PAGE 7

"3.5.1.1, first sentence: Delete and substitute "The portion of the case exposed to view from the front of the panel shall have a light-gray finish in accordance with formula No. 111, class 2 of MIL-E-15090."

PAGE 9

3.13: Delete and substitute:

"3.13 Voltage influence.- The voltage influence shall not exceed the value specified when tested in accordance with 4.6.8 (see 3.27)."

PAGE 10

3.24: Delete and substitute:

"3.24 Shock (high-impact).- When high-impact-shock resistant meters are tested in accordance with 4.6.19, no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, (except as specified herein), or unduly damaged. Minor cracking or distortion of pivots will be permissible only when conformance to the post shock performance has been demonstrated. Cracking of glass windows will be permissible only when readability of the complete scale is retained and no fragment has become dislodged. The permanent

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change in indication based on calibration made after resetting the pointer to zero or null position (if applicable) shall not exceed the value specified (see 3.27)."

PAGES 10 AND 11

3.25.1, line 6: Delete "Specification MIL-I-15024" and substitute "Standard MIL-STD-130".

PAGES 11 AND 12

3.27(d): Under "Limits" delete "3" and substitute "5".

3.27(f) Under "Limits": Delete and substitute:

"Potential system 8 volt amperes maximum per element at 120 volts 60 and 400 cycles for nominal 120 volt rating. For higher nominal voltage ratings the maximum volt ampere burden shall be increased proportionately. Current system 2.5 volts ampere maximum per element for 60 cycles and 16 volts ampere maximum per element of 400 cycles with current applied in phase with the voltage to produce end scale deflection while the potential system is energized at rated voltage and frequency. In a 2 element system, this is determined with all current coils similarly energized."

3.27(i): Under "Limits": Delete and substitute:

"1 percent maximum from 1.0 to 0.5 lagging power factor."

Add as paragraph 3.30:

"3.30 Power factor influence.- The power factor influence shall not exceed the value specified (see 3.27 and 4.6.21)."

4.1(b): Delete and substitute:

"(b) Quality conformance inspection (see 4.4)."

PAGE 13

Add as paragraph 4.2.11:

"4.2.11 Polyphase wattmeters.- Polyphase wattmeters may be connected as single phase wattmeters and subjected to the applicable tests specified herein subject to meeting the following requirements.

(a) No mutual action shall exist between elements of the wattmeter. This requirement shall be checked by energizing the potential circuit of one element and the current circuit of the other element. Deflection of the pointer shall not exceed plus or minus 0.5 percent of scale length for this connection.
(b) Both elements of the wattmeter shall be matched in their characteristics. This shall be checked by connecting the potential circuits in parallel and the current circuits in series opposing and energizing the wattmeter. Deflection of the pointer shall not exceed plus or minus 0.5 percent of scale length."

4.3: Delete and substitute:

"4.3 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. These tests shall consist of the tests specified in 4.6."

4.3.1, line 5: Delete "of 0.065 ± 0.010-ohm resistance".

Table I: Under "Specimens, Type designation", line 1: Delete "MR491W500ACWWA" and substitute "MR491W500ACWWH".

Table I: Under "Specimens, Type designation", line 2: Delete "MR492W1000ACWWA" and substitute "MR492W1000ACWWH".

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Table II: Under "Test" column: Delete "Frequency range" and substitute "Voltage influence."

Table II: Under "Test" column: Add "Power factor influence", under "Requirement paragraph" column, add "3.30"; and under "Test paragraph" column, add "4.6.21".

PAGES 15 AND 16

4.4 through 4.5.3 and Tables III, IV and V: Delete and substitute:

"4.4 Quality conformance inspection.- All wattmeters shall be subjected to the quality conformance inspection tests shown in table III. In case of rejection of a wattmeter to pass the quality conformance inspection, corrective measures may be approved by the Government inspector for up to three failures. The Naval Ship Engineering Center shall be advised. Over three failures shall be referred to the Naval Ship Engineering Center for action.

Table III - Quality conformance inspection tests.

Test	Requirement paragraph	Test paragraph
Visual and mechanical inspection (without opening case)	(See table VI)	4.6.1
Position influence	3.8	4.6.3
Initial accuracy (2 points)	3.9	4.6.4.2
Damping factor	3.10	4.6.5
Dielectric stress	3.20	4.6.15

PAGES 16 AND 17

Table VI, column 3: Delete.

PAGE 17

4.6.5, fourth sentence: Delete.

4.6.8: Delete and substitute:

"4.6.8 Voltage influence.- The voltage across the potential circuit shall be varied plus and minus 10 percent of rated voltage and a check for accuracy made at 0.8 power factor lagging and unity power factor. The number of scale readings shall be as specified in 4.2.9 and the readings compared with the corresponding readings of the standard meter."

PAGE 20

Add as paragraph 4.6.21:

"4.6.21 Power factor influence.- Readings shall be taken when the power factor is varied from 0.5, 0.8 lagging and unity power factor. The number of readings at each power factor shall be as specified in 4.2.9 and the readings compared with the corresponding readings of the standard meter."

4.7: Delete and substitute:

"4.7 Responsibility for inspection.- Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

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PAGE 21

6.3: Delete and substitute:

"6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-19088, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Ship Engineering Center, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

"6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, Pennsylvania, PA 19120."

PAGE 22

Patent notice: Delete.

PAGE 23

Figure 1: Delete dimension "1.250 $\pm$ .375" and substitute "0.625 $\pm$ 0.125".

NOTE: The margins of this amendment are marked "#" to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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